

Innovation-Aware LLMs (InnoLM)

CS6485: Capstone Project I

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Motivation

- Current models mostly store facts and struggle to understand how scientific ideas develop over time.
- In real research, new ideas build on older ones and follow a clear sequence of progress.
- Future models should move beyond memorization and learn how technologies grow and transform.
- Long chains of innovation are still difficult to process because of limited context capacity.
- **Example of idea evolution:** Attention → Transformer → Large Language Models.

Project Objective

- Instead of only aiming for correct outputs, the model should learn to reason about how ideas develop and change over time.
- This helps the model understand innovation as a process, not just a collection of facts.
- To support large models, training should be spread across multiple GPUs by splitting the model parameters efficiently.

Build a generative graph-language model (InnoLM) using the GOFA (Generative One-For-All) framework to unify GNNs and LLMs.

Literature Survey - Scientific LLMs

- **SciLitLLM:** Adapts LLMs for literature understanding via Continual Pre-training (CPT) on 625,000 academic papers and Supervised Fine-Tuning (SFT) [1]
- **The AI Scientist:** A comprehensive system designed for fully automated, open-ended scientific discovery, including idea generation and peer review [2].
- **Agent Laboratory:** Uses LLM agents as research assistants to transform human ideas into code and reports, utilizing iterative feedback [3].

[1] <https://arxiv.org/abs/2408.15545>

[2] <https://arxiv.org/abs/2408.06292>

[3] <https://arxiv.org/abs/2501.04227>

- **Global Summarization:** GraphRAG uses knowledge graphs and community detection (Leiden's algorithm) to handle queries requiring information from diverse text pieces [8] .
- **Think-on-Graph (ToG):** Performs iterative beam searches on KGs, enabling smaller LLMs to outperform larger models in specific reasoning tasks [9] .
- **Plan-on-Graph (PoG):** Decomposes complex queries into sub-objectives and navigates graphs with self-correcting mechanisms [10] .

[8] <https://arxiv.org/abs/2404.16130>

[9] <https://arxiv.org/abs/2307.07697>

[10] <https://arxiv.org/abs/2307.07697>

Literature Survey - Temporal & Interpretability

- **ChronoSense:** A benchmark for evaluating temporal understanding, focusing on Allen's interval relations (before, after, during) [4].
- **Graph Hawkes Transformer:** Architecture designed to capture structural and temporal dependencies using snapshots and Temporal Transformers [7].
- **TKE (Temporal Knowledge Editing):** Represents knowledge as time-scoped quadruples (s, r, o, t_s, t_u) to maintain historical facts while injecting new data [5].

[4] <https://arxiv.org/abs/2501.03040>

[7] <https://aclanthology.org/2022.emnlp-main.507.pdf>

[5] <https://arxiv.org/abs/2312.05497>

Model Architecture - GOFA Framework

- **Generative One-For-All (GOFA):** A unified model designed for joint graph-language modeling [6].
- **TAG Processing:** Specifically handles Text Attributed Graphs where each node represents a research paper with metadata (title, abstract, authors).
- **Dual-Component Design:**
 - **LLM Compressor:** Encodes textual information alongside structural connectivity
 - **LLM Decoder:** Autoregressively generates text responses based on compressed graph embeddings.
- **Optimization Goal:** Maximize the likelihood of generating the target text sequence given a graph context

GOFA Architecture

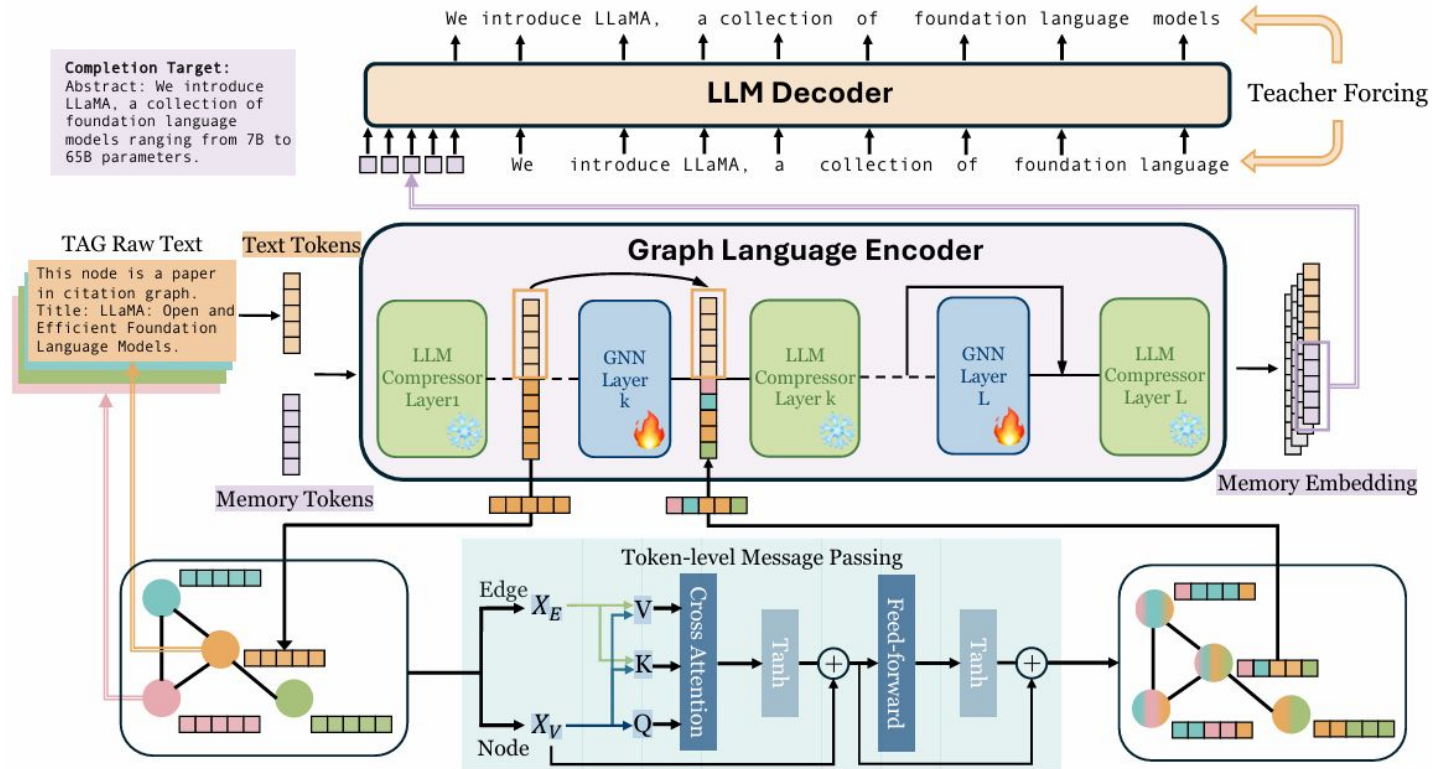


Fig 1. GOFA Architecture

LLM Compressor & GNN Interleaving

- **Context Compression:** Utilizes an In-Context Autoencoder (ICAE) to compress long node texts into K compact memory tokens.
- **Message Passing Architecture:** ICAE layers are interleaved with Graph Neural Network (GNN) layers [11].
- **Hybrid Learning:**
 - ICAE weights are frozen to preserve linguistic knowledge.
 - GNN weights are trained to capture structural relationships between papers.
- **Memory Update:** Memory tokens facilitate communication between textual embeddings and the graph structure during token-level message passing

ICAE Architecture

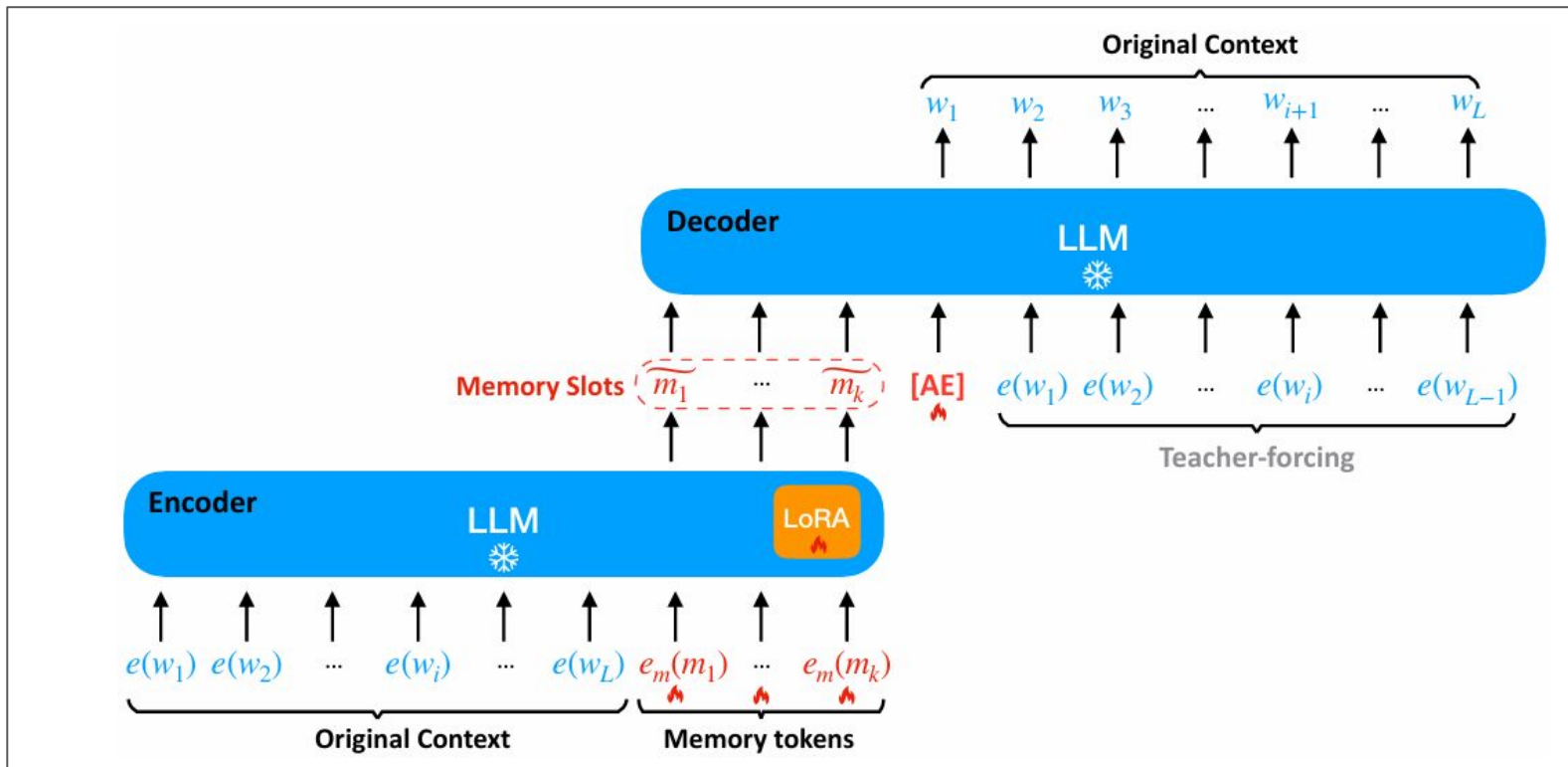


Fig 2. ICAE

Implementation Progress - Hardware Constraints

- **The Resource Gap:** Original GOFA training required 4x A100 (80GB VRAM).
- **Out-of-Memory (OOM) Challenges:** Training 8.5B parameter models with standard strategies (ZeRO-2) replicates model parameters across all GPUs, exceeding capacity.
- **Failed Attempt - Model Scaling:** Training a smaller 4B Mistral-based ICAE led to loss plateaus that failed to match 7B performance.
- **Failed Attempt - Quantization:** 8-bit quantization disrupted fine-grained weight updates from LoRA, leading to “gibberish” responses and low BLEU scores.

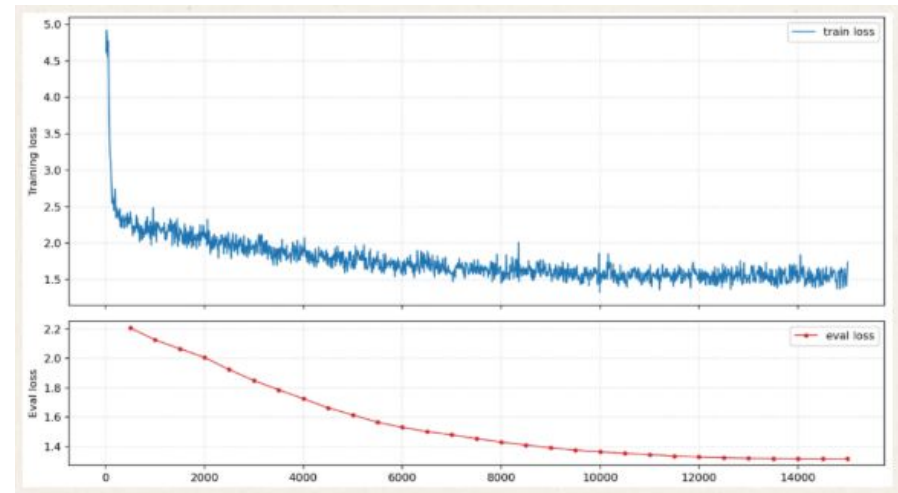


Fig 3. Mistral 4B vs. 7B: Training and Evaluation Loss Plateau Analysis

Optimized Sharding with FSDP

- **Strategy Shift:** Replaced DeepSpeed with PyTorch's Fully Sharded Data Parallel (FSDP).
- **Parameter Sharding:** Unlike ZeRO-2, FSDP shards model parameters, gradients, and optimizer states to save GPU memory.
- **Technical Hurdle:** FSDP's default behavior struggled with mixed frozen (ICAE) and trainable (LoRA) parameters.
- **The Solution:** Upgraded to latest PyTorch versions and utilized `use_orig_params = True` to enable computation on original unsharded parameters gathered via All-Gather operations.

FSDP Architecture

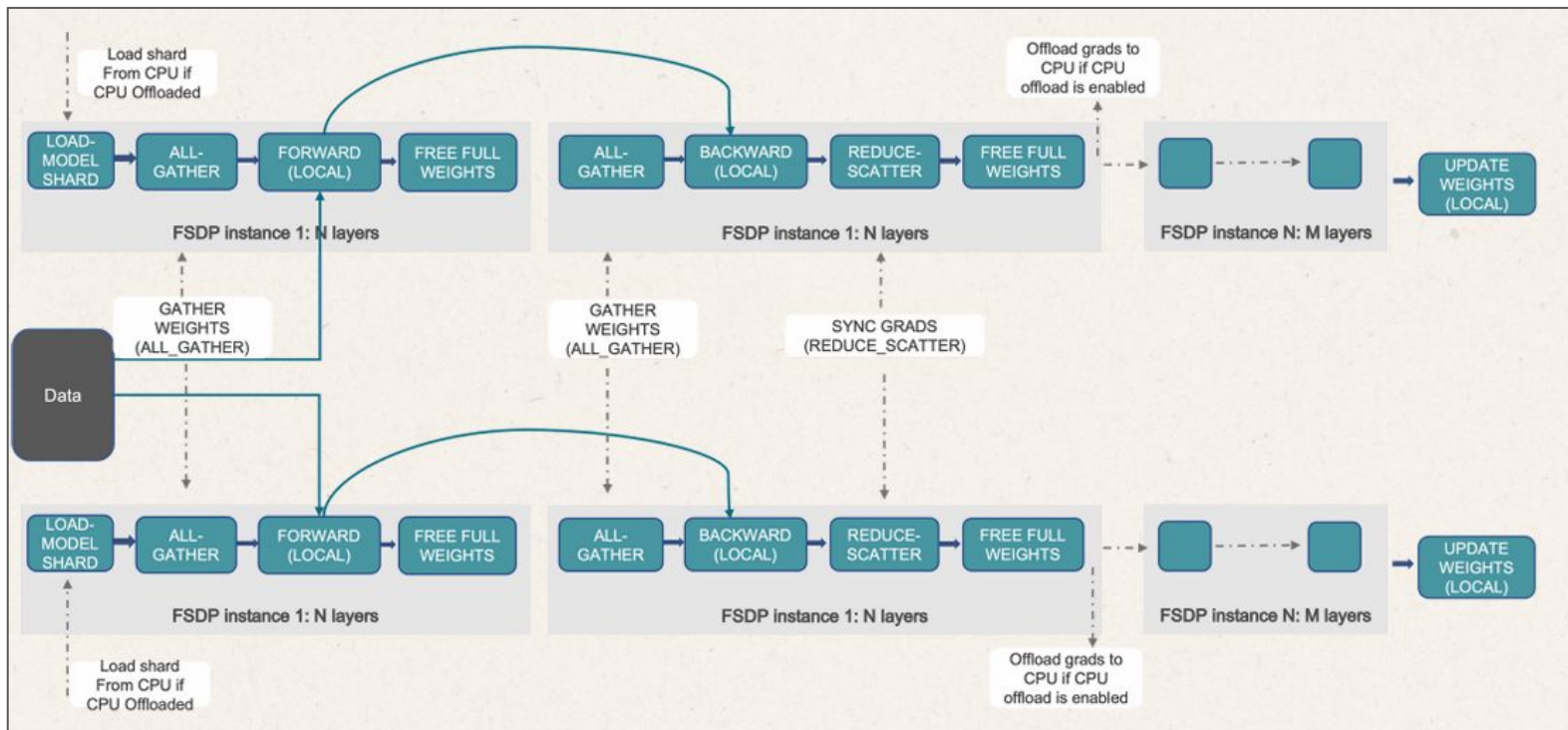


Fig 4. FSDP Architecture

Data Pipeline - SciEvo Processing

- **Dataset:** SciEvo(≈ 2 M papers); experiments use filtered citation-connected subsets for feasibility.
- **Input Construction:**
 - Retrieve all cited papers for a target paper p.
 - Summarize cited abstracts into 2-3 sentences using Phi-4-mini to reduce token overhead.
- **Target:** Predict the full abstract of paper p based on its title and summarized citations.
- **Optimization:** Used Hash-maps to store content indexed by Paper IDs for efficient prompt construction.

Experimental Results - Performance Metrics

- **Supervised Fine-Tuning (SFT):** Performed using LoRA to reduce computational overhead.
- **Quantitative Comparison (BLEU Scores):**

	Zero-shot	Few-shot	Finetuned
Full abstracts	0.021873923	0.000292668	0.026097466
Summarized abstracts	0.018845723	0.022436577	0.02946509

- **Analysis:** SFT outperformed non-finetuned settings across all configurations.

Training Dynamics

- **Loss Convergence:** Observed a steady decrease in average cross-entropy training loss across 12 epochs.
- **Performance Plateau:** Despite loss reduction, validation BLEU scores remained nearly constant (~ 0.207).
- **Hyperparameter Tuning:** Currently adjusting learning rates and gradient accumulation steps to optimize the loss curve.

Epoch	Training Loss	Validation BLEU
0	2.1963	0.207
1	2.0905	0.207
2	2.0699	0.208
3	2.0453	0.209
4	2.0032	0.210
5	1.9409	0.207
6	1.8727	0.208
7	1.8041	0.208
8	1.7432	0.209
9	1.6906	0.208
10	1.6438	0.207
11	1.5993	0.207
12	1.5564	0.207

Error Analysis & Technical Debugging

- **Jumbled Validation Outputs:** Initial FSDP runs produced incoherent text: *"None theound the-Dim Communitation"*.
- **Root Cause Diagnosis:** Softmax was likely occurring on sharded subsets of tokens because the output layer was sharded across GPUs.
- **Verification:** Constructed a sample graph JSON and ran using a single GPU (non-FSDP), which produced clean, non-jumbled outputs.
- **Mitigation:** Attempted to keep the last layer unsharded to resolve token distribution issues.

None theound the-Dim Communitation, Two-hard Perblems, theum Unuctions, Abstract: None quantum problem in quantum theory is the the-case instances to a-case ones. In are quantum classes that as N,, are such-case to average-case reductions, However, the other complexity complexity such as NP, the worst is far is that negative. and the sense that no average of worst reductions is imply aapses of the polynomial hierarchy.PH). Inasing oneographic primitives on such.g., one one-case hardness of onever

Fig 5. Jumbled output

[illegible]

Model: Qwen2.5-Omni-7B
BLEU Score : 0.0138

Future Work

- **Mechanistic Interpretability (MI):** Analyze InnoLM's internal reasoning using Causal Tracing on scientific tasks.
- **Neuronpedia:** Benchmark and visualize discovered features on the **Neuronpedia** platform to align with current global SAE research.
- **Feature Discovery:** Apply Sparse Autoencoders (SAEs) and ACDC to identify interpretable circuits and reduce polysemanticity
- **Concept Geometry:** Probe whether scientific facts and innovation trajectories follow linear or ordered structures in embedding space.

Thank You!